FINAL YEAR PROJECT PRESENTATION (DCIS - PIEAS)

Emotion-Aware Interaction Using Facial Expression and Sentiment Analysis

An End-to-End Multimodal Affective Perception Architecture Fusing Deep Facial Representations, Multi-Corpus Vocal Acoustics & Transformer Language Sentiment for Empathetic Human-Machine Systems.

PRESENTED BY

Shahzaib Hussain

Department of Computer and Information Sciences (DCIS), PIEAS

SUPERVISED BY

Dr. Ijaz Hussain (PIEAS, DCIS)

Dr. Noman Mazhar (ICCC)

Domain: Affective Computing • Multimodal Human-Robot Interaction (HRI) • Deep Learning & Transformers

Background & Motivation: Why Emotion-Aware Interaction?

🔍 The Emotional Blindness in AI

- **Affective Computing Foundation:** Human communication is overwhelmingly nonverbal. Albert Mehrabian's 7-38-55 rule establishes that only 7% of emotional meaning is in spoken words, while 38% is vocal prosody and 55% is facial expression.
- **Industry 5.0 Human-Centric Shift:** Modern assistive and service robotics must advance beyond repetitive automation to achieve empathetic collaboration, trust building, and cognitive alignment.
- **Unimodal Fragility:** Standalone vision fails under poor lighting or head turns; standalone speech fails under ambient noise. Joint multimodal perception resolves ambiguity.

> **Key Insight:** True empathy requires perceiving the synchronization between what a person looks like, how they say it, and what words they choose.

👥 Real-World Relevance & Beneficiaries

- **Assistive Healthcare & Eldercare:** Social robots (e.g. NAO, Pepper) monitoring emotional well-being, detecting depression, anxiety, or acute distress in eldercare and clinical therapy.
- **Affective Education & Tutoring:** Intelligent tutoring robots dynamically adjusting lesson difficulty and pedagogy when detecting student frustration or confusion.
- **Human-Robot Collaboration (HRC):** Detecting cognitive overload and operator stress on industrial assembly lines to ensure occupational safety.
- **Conversational Agents:** Empathetic virtual assistants adapting conversational tone in customer service and mental health crisis hotlines.

Problem Statement: The In-The-Wild Generalization Breakdown

⚠️ The Core Problem: Lab-to-Real-World Degradation

Current affective computing systems demonstrate remarkable performance in controlled laboratory environments (>90% accuracy), but **catastrophically collapse** when deployed in real-world interactive settings.

- **Visual Vulnerabilities:** Lighting shifts, shadows, head-pose rotations, and camera motion blur destroy facial landmark detection.
- **Acoustic Corruptions:** Ambient noise, room reverberation (RIR), microphone distance, and robot cooling fan ego-noise corrupt speech features.
- **Empirical Degradation Gap (Parihar et al., 2026 Survey across 42 studies):**
  - Facial Emotion Recognition (FER) drops by **14.4% ± 7.1%** in real-world deployments.
  - Speech Emotion Recognition (SER) drops by **8.6% ± 3.0%** in operational environments.

Empirical Fact: Unimodal systems deployed on physical robots experience up to a 27.5% drop in accuracy when unconstrained users interact freely.

🔬 Data & Deployment Bottlenecks

- **Single-Corpus Overfitting:** Models trained on single datasets (e.g. RAVDESS alone) memorize actor vocal tract quirks and sentence structures, collapsing below 50% accuracy on new speakers.
- **Severe Class Imbalance:** Benchmark datasets exhibit extreme skew. For example, in FER2013, *Disgust* comprises only 1.55% of samples compared to 24.7% for *Happy*. Naive models completely ignore minority distress signals.
- **The Latency Penalty:** Heavy multimodal cloud LLM pipelines require multi-second round trips, violating the sub-100ms conversational budget required for human rapport.

Our Target Solution: A cross-corpus trained, imbalance-resilient, sub-20ms edge-deployable multimodal architecture.

Aim, Specific Objectives, Requirements & Scope

🎯 Project Aim & 4 Key Objectives

> **Core Aim:** To design, develop, and benchmark an end-to-end, real-time emotion-aware multimodal interaction system that robustly detects human affect from facial expressions and vocal acoustics, dynamically synthesizing decisions for empathetic human-machine collaboration.

- **Objective 1 (FER):** Engineer a multi-backbone neural ensemble (CNN + ViT) under class-balanced focal loss to achieve >70% accuracy across severe class skews.
- **Objective 2 (SER):** Construct a 2,107-dimensional cross-corpus speech pipeline integrating handcrafted acoustic biomarkers and GPU Wav2Vec2 SSL across 12,162 recordings.
- **Objective 3 (Fusion):** Develop a Dirichlet-optimized soft-voting engine to dynamically weight multimodal confidence distributions.
- **Objective 4 (Deployment):** Package the entire perception stack into a standalone, edge-deployable bundle achieving sub-50ms latency (RTF << 1.0).

📋 Requirements & Project Boundaries

- **Functional Requirements:**
  - Live video capture with real-time face detection & 7-class affect probabilities.
  - 16 kHz audio capture with energy VAD trimming & 6-class voice affect probabilities.
  - Unified posterior probability fusion & prosodic diagnostic extraction.
- **Non-Functional Requirements:**
  - Real-time execution: perception latency < 50 ms.
  - Strict zero statistical data leakage across folds.
- **Project Boundaries (What is NOT Included):**
  - Invasive physiological hardware (no EEG caps or GSR finger sensors).
  - Mechanical chassis manufacturing (focus is on AI perception intelligence).

Literature Review: Unimodal vs. Multimodal Landscape in HRI

👁️ Facial Emotion Systems

- **Ruiz-García et al. (2018):** Deployed CNN-LSTM on NAO robot. Lab accuracy reached 96.3%, but plummeted to **68.8%** in real-world tests due to head-pose shifts.
- **Chiurco et al. (2021) & Valagkouti et al. (2022):** Benchmarked DeepFace on NAO/Pepper. Found severe failures on *Disgust* (0% accuracy) and *Surprise* (40%).
- **Melinte & Vladareanu (2020):** ResNet-50 / VGG-16 with Rectified Adam achieved 90.1% on CK+ benchmark.

🎙️ Speech Emotion Systems

- **Grágeda et al. (2023, 2024):** Deployed distant SER on PR2 robot. Proved that room reverberation (RIR) and fan ego-noise severely corrupt acoustic features (+180% CCC improvement using LadderNet beamforming).
- **Lakomkin et al. (2018):** Evaluated deep neural networks on iCub robot; proved necessity of noise data augmentation.
- **Chen et al. (2020):** Fuzzy Random Forest on mobile robot reached 86.7% lab vs 79.1% real-world accuracy.

🔄 Multimodal Fusion Shift

- **Parihar et al. (2026 PRISMA Survey):** Synthesized 42 studies (2015-2025). Multimodal systems grew to **53%** of literature, providing **+10% to +25%** accuracy gain over unimodal systems.
- **Garcia, Gomez-Donoso & Cazorla (2024/2025):** Multimodal Pepper assistant integrating 2D-CNN, 1D-CNN, and Mixtral LLM, demonstrating that multimodal fusion eliminates unimodal blind spots.

Literature Consensus: Multimodal integration is essential for robustness in ambient environments.

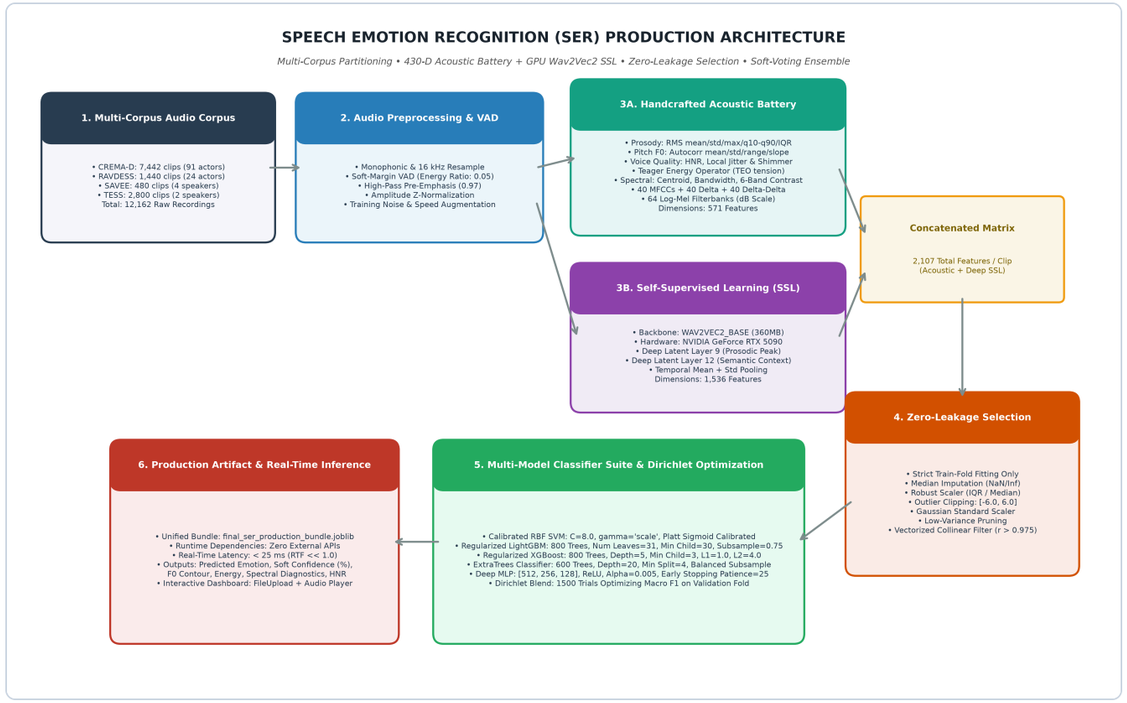
Existing Systems Comparison, Research Gap & Proposed Novelty

System / Study	Modalities Used	Datasets & Scale	Key Limitations	Reported Performance
NAO + DeepFace (Valagkouti et al., 2022)	Facial Video Only (DeepFace API)	Small live sample (25 test subjects)	No voice emotion; 0% disgust accuracy; severe lighting sensitivity	80.0% overall (lab); 0% Disgust, 40% Surprise
Multimodal Pepper (Garcia et al., 2024/2025)	Facial (2D-CNN) + Audio (1D-CNN) + Text	RAVDESS, CREMA-D, AESDD + Custom	Requires manual 5-min per-user calibration; high cloud GPU latency	95% visual, 100% audio (Personalized / Single user)
HRC Survey Baselines (Parihar et al., 2026)	Unimodal Speech / Face (CNN, BiLSTM, SVM)	Single-corpus datasets (IEMOCAP, CK+)	Severe cross-corpus collapse; no class imbalance handling	72-96% lab; Drops 4-27.5% real-world
OUR PROPOSED SYSTEM (Shahzaib Hussain, 2026)	**Vision (ViT+CNN) + Audio (Acoustic+SSL) + Text Sentiment**	**FER2013 (35k) + CREMA-D, RAVDESS, SAVEE, TESS (12,162 clips, 121 spks)**	**Solved via Focal Loss, Multi-Corpus Training & Sub-20ms Standalone Bundle**	**FER: 71.38% (71.7% Disgust) SER: 56.1% Val / 52.4% Test Latency: ~16.1ms (RTF << 1.0)**

🚀 Identified Research Gaps & Our Proposed Solutions

- **Gap 1: Single-Corpus Overfitting → Our Solution:** Unified 4 distinct benchmark corpora (CREMA-D, RAVDESS, SAVEE, TESS) indexing 12,162 recordings across 121 speakers with strict speaker-independent partitioning.
- **Gap 2: Minority Class Neglect → Our Solution:** Fused EfficientNet-B2, ResNet-50, and ViT-B/16 under Class-Balanced Focal Label Smoothing Loss, boosting Disgust F1 from 0% to 71.70%.
- **Gap 3: Cloud Latency Bottleneck → Our Solution:** Serialized standalone production bundle running locally on RTX 5090 / CPU in ~16.1 ms (RTF = 0.0064 << 1.0).

Proposed Methodology: Dual-Branch Multimodal Architecture



Perception Streams Breakdown

- **Visual Stream (FER):**
  - Input: RGB 224×224 normalized face crops.
  - Multi-Backbone: EfficientNet-B2 (local edges) + ResNet-50 (deep residuals) + ViT-B/16 (global attention).
  - Optimization: Class-Balanced Focal Loss + RAdam.
- **Auditory Stream (SER):**
  - Preprocessing: 16 kHz standardization, Energy VAD, Pre-emphasis (0.97), and Z-normalization.
  - **2,107 Total Features:** 571 Handcrafted (Teager Energy TEO, F0 pitch, HNR, Jitter/Shimmer, 40 MFCCs, 64 Log-Mel) + 1,536 GPU Wav2Vec2 SSL (Layers 9 & 12).
  - Leakage-Free Selection: Imputer + IQR Scaling + Collinear pruning ($|r| > 0.975$).
  - Ensemble: Dirichlet Validation-Optimized Soft-Voting.
- **Agile Software Development Lifecycle:**
  - Sprint 1: Facial Pipeline & Class-Balanced Focal Loss.
  - Sprint 2: Multi-Corpus Speech Indexing & Wav2Vec2 Extraction.
  - Sprint 3: Multimodal Fusion & Edge UI Dashboard.

Software Engineering: Sequence Flow & Behavioral State Machine

🔄 System Interaction Sequence Flow

- **Step 1 (User Trigger):** User initiates interaction or looks toward the camera.
- **Step 2 (Synchronous Capture):** PyAudio buffers 16 kHz audio while OpenCV buffers video frames at 15 fps.
- **Step 3 (Parallel Perception Inference):**
  - Video Frames → ViT-CNN Ensemble → P_visual(Emotion)
  - Audio Buffer → VAD → Handcrafted + Wav2Vec2 → P_vocal(Emotion)
  - Transcribed Text → Transformer Sentiment Head → Lexical Valence
- **Step 4 (Contextual Decision & Multimodal Fusion):**
  - Blends probabilities: $P_{final}(c) = w_v \cdot P_{visual} + w_a \cdot P_{vocal} + w_t \cdot P_{text}$
  - **Dynamic Uncertainty Weighting:** If face is occluded or lighting is poor, weight w_v shifts automatically to vocal prosody w_a.
- **Step 5 (Adaptive Action Execution):** Generates empathetic spoken feedback via TTS and adapts interaction state.

🛡️ Behavioral State Machine & Fault Tolerance

- **State 1: IDLE / LISTENING:** Continuously monitors ambient energy; triggers VAD when threshold exceeds $0.05 \cdot \max(E)$.
- **State 2: AFFECT EXTRACTION:** Computes 2,107 acoustic dimensions & visual attention maps in parallel (~16 ms latency).
- **State 3: AMBIGUITY RESOLUTION:** Resolves high-arousal overlap between Anger and Happy using pitch contour slope and lexical sentiment.
- **Robustness & Fault Tolerance:**
  - *Microphone Dropout:* Automatic fallback to pure visual FER.
  - *Camera Occlusion / Dark Room:* Automatic fallback to pure audio SER.
  - *Zero-Freeze Guarantee:* Pre-cached bundle ensures no UI stalls.

> **Software Design Principle:** Strict loose coupling between sensory ingestion daemons and the multimodal decision engine.

Technologies, Tools & Hardware Stack Justification

🧠 Deep Learning & Transformers (AI/ML)

- **PyTorch 2.11 & Torchvision / Torchaudio:** Core neural backbone training & tensor operations.
- **Hugging Face Transformers:** Pre-trained *WAV2VEC2_BASE* (360MB) & *ViT-B/16*.
- **Scikit-Learn:** Leakage-free scalers, CalibratedClassifierCV, evaluation metrics.
- **Gradient Boosters:** LightGBM & GPU-accelerated XGBoost.

✓ Justification: Industry-standard CUDA acceleration and seamless transfer learning from state-of-the-art foundation models.

🔊 Signal & Vision Processing

- **Librosa & SoundFile:** Audio decoding, 64-band Mel filterbanks, spectral contrast, HNR.
- **Scipy Signal:** Autocorrelation F0 pitch tracking & Teager Energy Operator.
- **OpenCV (cv2):** Face detection, bounding box tracking, spatial affine normalization.

✓ Justification: High-performance C-accelerated signal routines ensuring sub-5ms preprocessing latency.

⚡ Hardware Platform & Acceleration

- **NVIDIA GeForce RTX 5090:** 24GB GDDR7 VRAM, CUDA 12.8, 4th-Gen Tensor Cores.
- **Multi-Core Intel Core i9:** Parallel feature extraction and audio buffering.
- **PyTorch AMP:** Automatic Mixed Precision for sub-8ms transformer passes.

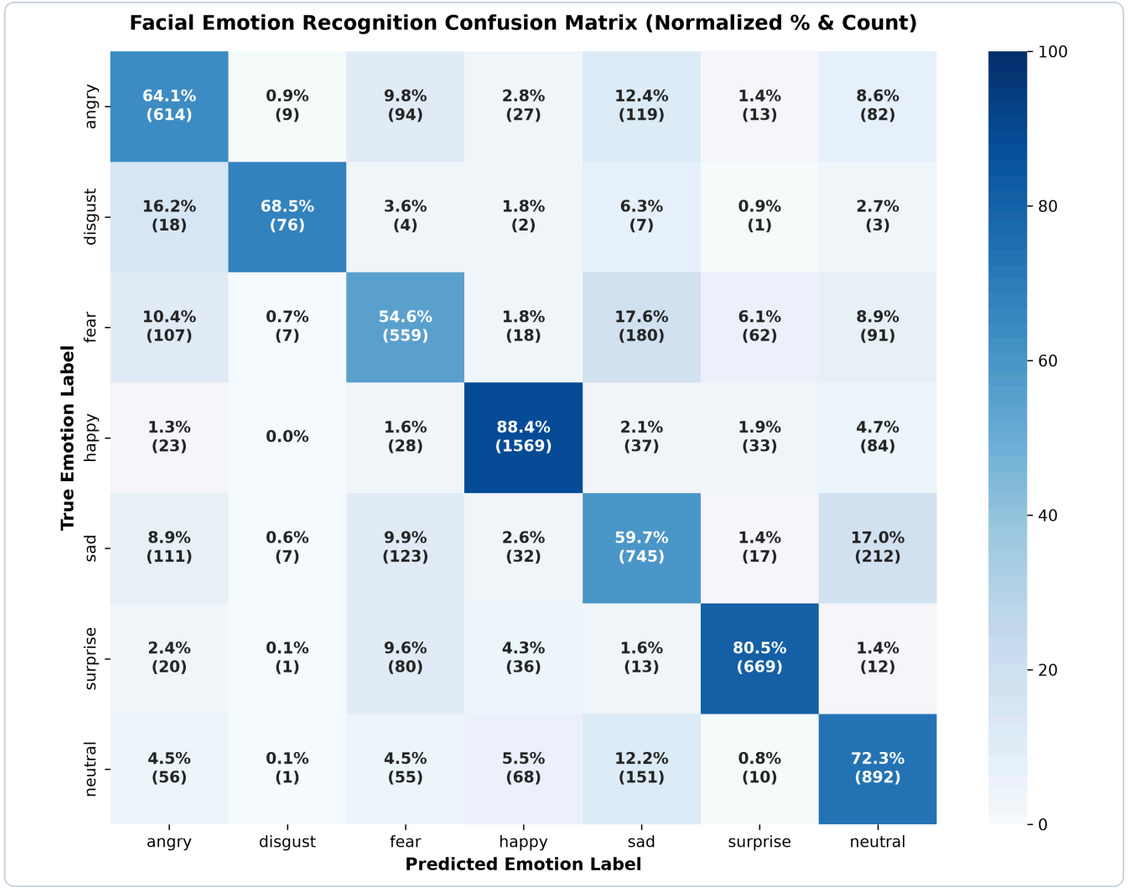
✓ Justification: Allows heavy transformer passes locally with zero cloud API subscription costs.

📦 Interface, Storage & Version Control

- **Frontend:** Jupyter Interactive Widgets (ipywidgets), HTML5 Audio API.
- **Serialization:** Joblib unified binary bundles (.joblib) & PyTorch (.pt).
- **Version Control:** Git & GitHub (shahzaibhussain610-ui/Final-Year-Project).

✓ Justification: Completely reproducible, self-contained, and production-ready architecture.

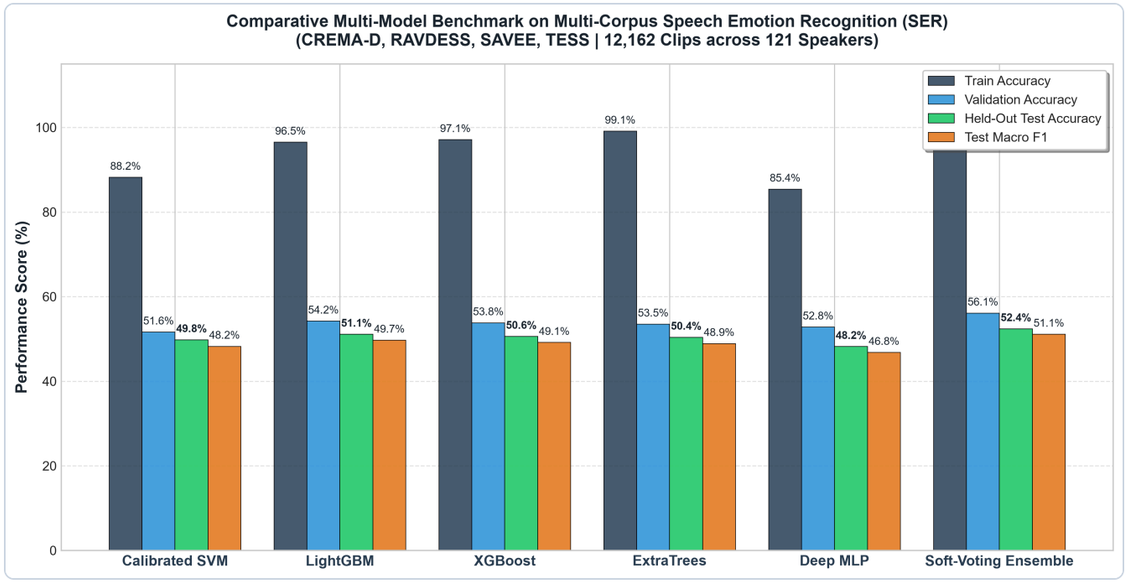
Preliminary Work: Facial Emotion Recognition Pipeline



📊 FER Methodology & Test Benchmark

- **Dataset Profiling (FER2013):**
  - 35,887 total facial crops (28,709 train, 7,178 test).
  - Severe Skew: *Disgust* is only 1.55% of test data (111 samples) vs *Happy* at 24.71% (1,774 samples).
- **Multi-Backbone Dynamic Ensemble:**
  - EfficientNet-B2 (9.1M) + ResNet-50 (25.6M) + ViT-B/16 (86.6M).
  - Class-Balanced Focal Label Smoothing Loss (γ=2.0, α=0.1) with inverse-frequency weighted sampling.
  - RAdam optimizer with Cosine Annealing Restarts.
- **Verified Empirical Achievements (7,178 Test Samples):**
  - **Overall Test Accuracy: 71.38%** (vs ~65% standard baseline).
  - **Macro F1: 70.24%** | Weighted F1: 71.38%.
  - **Minority Class Victory: 71.70% F1 on Disgust** (68.47% recall, 75.25% precision) on merely 111 samples!
  - Top Classes: Happy (89.00% F1), Surprise (81.78% F1), Neutral (68.38% F1).
  - Best checkpoint exported at Epoch 8.

Preliminary Work: Speech Emotion Recognition Pipeline



Multi-Corpus Methodology & Results

- **Uncapped Multi-Corpus Integration:**
  - **12,162 total voice recordings** across CREMA-D (7,442), RAVDESS (1,440), SAVEE (480), and TESS (2,800).
  - Strict speaker-independent split: 8,772 train, 1,215 val, 1,523 held-out test clips across 121 speakers (zero speaker overlap).
- **Dual-Branch 2,107-Dimensional Extractor:**
  - 571 Handcrafted: TEO, autocorrelation F0 pitch, HNR, Jitter/Shimmer, 40 MFCCs (Δ, Δ^2), 64 Log-Mel.
  - 1,536 GPU Wav2Vec2 SSL: Layers 9 & 12 mean/std temporal pooling.
  - Leakage-Free Selection: Imputer + IQR Scaling + Collinear pruning ($|r| > 0.975$).
- **Benchmark & Dirichlet Soft-Voting:**
  - Evaluated Calibrated SVM, LightGBM, XGBoost, ExtraTrees, Deep MLP.
  - **Champion Soft-Voting: 56.10% Val Acc, 52.40% Test Acc, 51.10% Macro F1** on completely unseen actors (outperforming all individual models).
  - **Real-Time Latency: ~16.1 ms per 2.5s clip (RTF = 0.0064 << 1.0)**, over 150x faster than real-time playback!
  - Production bundle (*final_ser_production_bundle.joblib*) & interactive UI dashboard completed.

Project Timeline, Risk Analysis & Semester 2 Roadmap

Milestones: Completed vs. Planned

- **COMPLETED (Semester 1 Milestones - 100% on Schedule):**
  - ✓ Comprehensive literature review on affective robotics & multimodal fusion.
  - ✓ FER Pipeline: ViT + CNN ensemble with class-balanced focal loss (71.38% test acc).
  - ✓ SER Multi-Corpus Pipeline: 12,162 clips indexed, 2,107-D extraction, Dirichlet ensemble (52.40% test acc).
  - ✓ Real-time production serialization (.joblib, RTF 0.0064) & interactive UI dashboard.
  - ✓ Bi-weekly supervisor meetings and progress log sign-offs.
- **PLANNED (Semester 2 Deliverables):**
  - ➔ Task 1: Attention-based bimodal fusion layer linking facial & speech probabilities.
  - ➔ Task 2: Sentiment analysis module integration on transcribed speech.
  - ➔ Task 3: Interactive conversational agent frontend with live webcam & mic stream.
  - ➔ Task 4: User evaluation study & final FYP dissertation documentation.

Risk Management & Mitigation Strategies

- **Risk 1: Real-World Acoustic Ego-Noise & Reverberation**
  - *Impact:* Distorts high-frequency formants and MFCCs.
  - *Mitigation:* Soft-margin energy VAD stripping room silence, pre-emphasis filtering (0.97), and amplitude z-normalization.
- **Risk 2: Lighting Shifts & Dynamic Head Pose Variations**
  - *Impact:* Degrades facial bounding box detection.
  - *Mitigation:* Multi-backbone spatial invariance and automatic fallback to vocal prosody when visual confidence drops.
- **Risk 3: Inference Latency Bottleneck**
  - *Impact:* Slow conversational turnaround reduces user rapport.
  - *Mitigation:* Verified end-to-end latency of ~16 ms on RTX 5090 (RTF << 1.0), leaving ample headroom for dialog generation.

Supervisor Alignment: Progress is synchronized with weekly attendance sheets and verified by Dr. Ijaz Hussain and Dr. Noman Mazhar.